

POTTSTOWN, PA, USA

WMO: 725109

Lat: 40.238N

Lon: 75.555W

Elev: 288

StdP: 14.54

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 54782

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.4	13.4	-5.6	4.1	11.6	-0.5	5.4	16.6	21.2	24.5	19.6	26.8	6.3	300	0.422

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	91.4	74.2	89.0	72.8	86.4	72.0	76.9	86.7	75.5	83.9	74.4	82.2	7.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.1	128.7	80.7	73.0	123.7	79.2	72.0	119.7	78.4	40.4	86.8	39.1	84.4	37.9	82.4	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	16.2	13.4	3.8	94.8	4.9	4.0	0.3	97.7	-2.6	100.0	-5.4	102.2	-9.0	105.1
			2.4	79.5	4.6	1.2	-0.9	80.4	-3.6	81.1	-6.2	81.8	-9.5	82.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.6	30.6	32.6	41.1	52.6	62.5	70.5	75.6	73.5	66.8	55.2	44.6	36.0	
	DBStd	17.59	10.12	9.40	9.60	8.95	8.02	6.23	4.95	5.04	6.42	8.31	8.48	8.77	
	HDD50	2167	606	493	307	69	4	0	0	0	0	41	204	443	
	HDD65	5158	1067	908	742	386	147	22	0	3	54	320	611	898	
	CDD50	3476	3	5	30	147	392	616	794	728	505	203	43	10	
	CDD65	993	0	0	0	14	70	188	329	266	109	17	0	0	
	CDH74	8815	0	0	10	192	671	1673	3103	2216	839	109	2	0	
	CDH80	2936	0	0	0	52	190	523	1207	724	224	16	0	0	
Wind	WSAvg	4.9	6.0	6.2	6.5	6.1	4.4	4.1	3.7	3.2	3.7	4.4	5.0	5.5	
Precipitation	PrecAvg	46.4	3.4	2.6	4.2	3.7	3.6	4.3	4.5	4.1	4.6	4.1	3.4	4.1	
	PrecMax	61.7	6.3	6.1	7.1	7.2	7.1	10.4	9.3	9.1	11.7	10.0	5.3	8.6	
	PrecMin	33.5	1.4	0.6	0.9	1.6	1.0	1.7	0.6	1.3	0.5	0.2	1.1	0.8	
	PrecStd	6.9	1.5	1.2	1.5	1.6	1.4	2.2	2.0	1.7	2.8	2.5	1.2	1.8	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.7	66.4	74.4	86.3	89.0	92.8	96.5	93.5	90.3	81.8	71.4	65.7	
		MCWB	54.4	56.1	59.2	65.9	71.2	74.5	74.8	74.7	72.2	68.1	61.6	61.0	
	2%	DB	56.0	58.4	66.9	79.8	85.1	88.9	92.1	90.1	86.0	76.9	66.6	59.4	
		MCWB	50.7	50.5	55.1	62.0	69.8	72.5	74.8	73.9	71.5	66.8	58.9	55.9	
	5%	DB	49.7	51.5	61.7	73.8	81.5	85.8	89.6	86.6	82.0	73.0	63.2	54.2	
		MCWB	44.6	44.2	51.8	58.8	67.8	71.7	73.5	72.7	68.5	64.2	56.1	50.2	
	10%	DB	45.1	46.0	56.6	68.9	77.3	82.7	86.7	83.9	79.0	69.5	58.6	49.3	
		MCWB	40.6	39.7	48.5	56.4	65.2	69.6	72.6	71.3	67.6	62.3	52.2	44.9	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.7	59.5	61.8	68.6	74.3	76.8	79.1	78.7	76.1	72.2	64.4	61.5
			MCDB	60.7	64.9	71.4	80.1	85.5	88.8	91.1	88.9	84.6	76.5	68.5	64.1
2%		WB	51.1	51.2	57.7	64.6	71.8	75.1	77.2	76.6	74.1	69.4	60.9	56.4	
		MCDB	54.9	57.0	64.6	75.7	81.2	84.5	87.7	84.9	80.8	74.4	65.6	58.8	
5%		WB	45.7	44.4	53.3	61.6	69.6	73.4	75.8	75.1	72.4	66.7	57.3	51.0	
		MCDB	49.4	49.9	59.7	70.7	78.9	81.9	84.7	82.4	78.0	71.9	61.7	53.6	
10%		WB	40.9	40.5	48.7	58.4	67.5	71.9	74.6	73.6	70.4	63.2	54.0	45.7	
		MCDB	44.8	45.5	56.1	67.0	74.7	79.7	82.8	80.4	76.0	67.1	57.4	48.2	
Mean Daily Temperature Range	5% DB	MDBR	14.9	16.4	18.4	20.6	20.0	18.9	18.7	18.3	19.4	18.6	17.9	14.8	
		MCDBR	22.4	23.9	25.9	28.0	24.4	22.1	21.8	20.9	22.9	22.4	23.0	20.3	
		MCWBR	17.8	17.6	17.0	14.2	11.5	9.6	8.3	8.7	10.6	12.6	15.9	17.5	
	5% WB	MCDBR	20.4	21.9	23.1	23.6	21.5	18.7	17.9	16.8	17.1	17.6	18.6	18.4	
		MCWBR	17.5	17.8	17.6	14.5	11.1	9.4	7.6	8.0	9.7	11.5	15.1	18.1	
Clear-Sky Solar Irradiance	taub	0.308	0.322	0.359	0.395	0.438	0.501	0.499	0.497	0.409	0.360	0.325	0.307		
	taud	2.450	2.389	2.362	2.288	2.198	2.062	2.083	2.073	2.290	2.441	2.504	2.498		
	Ebn at Noon	272	283	283	279	268	250	249	246	263	266	264	263		
	Edh at Noon	25	31	35	41	45	52	51	50	38	29	24	23		
All-Sky Solar Radiation	RadAvg	590	877	1158	1505	1683	1838	1882	1636	1352	957	674	506		
	RadStd	51	99	90	138	201	149	93	103	165	102	61	55		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (72 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+221	+126

Nomenclature: See separate page